

17. For all sets A and B , if $A - B = \emptyset$, then $B = \emptyset$

The disproof of this would show that $A - B = \emptyset$ and $B \neq \emptyset$. Consider when $A = \{1, 3\}$ and $B = A$, then $A - B = \emptyset$ and $B \neq \emptyset$. So the statement is disproven.

30. There exist integers a and b for which $42a + 7b = 1$.

To disprove this with contradiction let's say this is true. $7 \mid 42a$ and $7 \mid 7b$ means 7 will always be a common divisor of there sum since $42a + 7b = 7(6a + b)$ can be written as $7(\text{some integer})$ means only the fraction $\frac{1}{7}$ would result in 1.